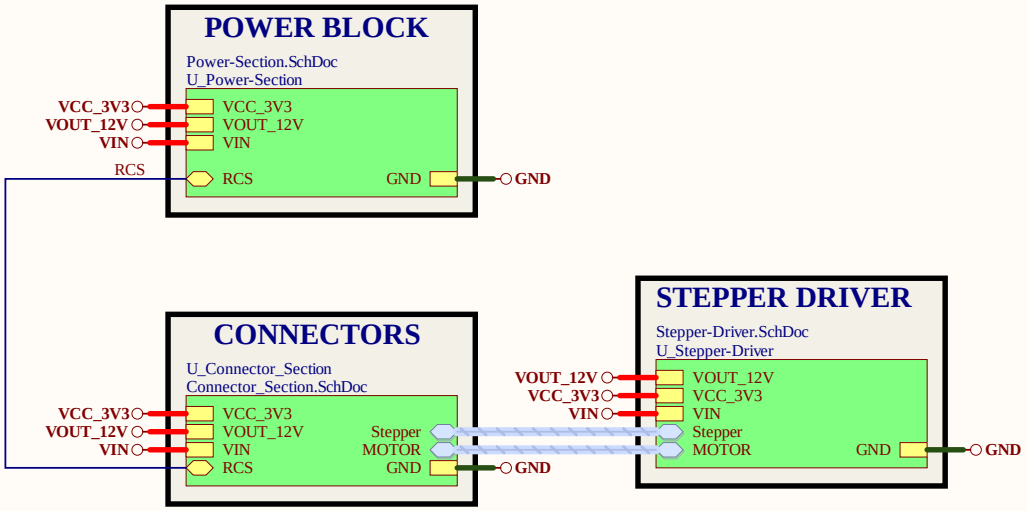
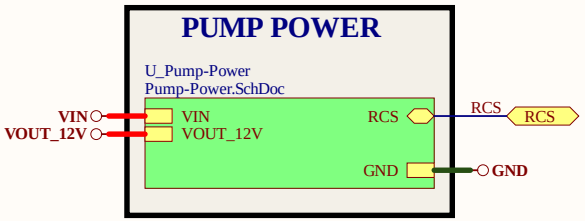
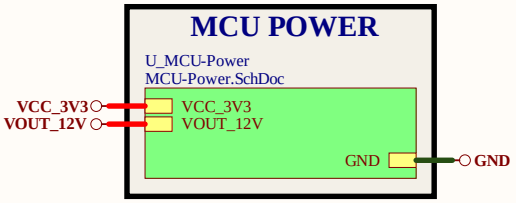
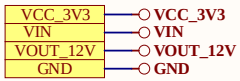
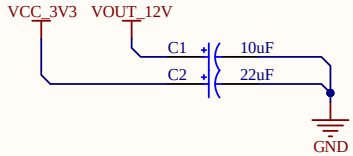

POWER BLOCK
Power-Section.SchDoc
U_Power-Section
VCC_3V3
VOUT_12V
VIN
RCS
VCC_3V3
VOUT_12V
VIN
RCS
GND
GND
CONNECTORS
U_Connector_Section
Connector_Section.SchDoc
VCC_3V3
VOUT_12V
VIN
RCS
VCC_3V3
VOUT_12V
VIN
RCS
Stepper
MOTOR
GND
GND
STEPPER DRIVER
Stepper-Driver.SchDoc
U_Stepper-Driver
VOUT_12V
VCC_3V3
VIN
VOUT_12V
VCC_3V3
VIN
Stepper
MOTOR
GND
GND


PUMP POWER
U_Pump-Power
Pump-Power.SchDoc
VIN
VOUT_12V
VIN
VOUT_12V
RCS
GND
RCS
RCS
GND
GND

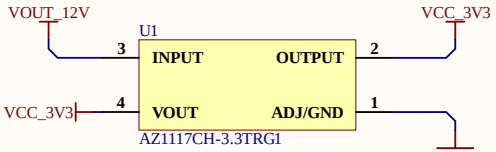

MCU POWER
U_MCU-Power
MCU-Power.SchDoc
VCC_3V3
VOUT_12V
VCC_3V3
VOUT_12V
GND
GND


VCC_3V3
VIN
VOUT_12V
GND
VCC_3V3
VIN
VOUT_12V
GND

DECOUPLING CAPACITORS

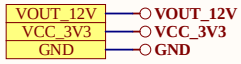


3V3 REGULATOR

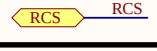


Fixed regulator.
VOUT= 3V3/1A

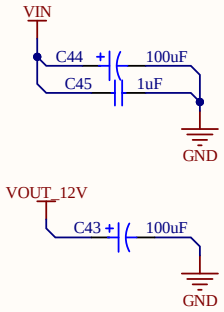
POWER SIGNALS



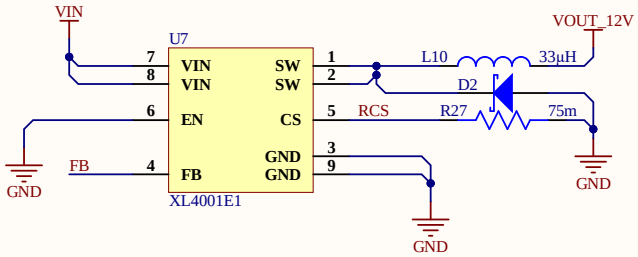
CURRENT SENSE INPUT



DECOUPLING CAPACITORS



12V BUCK REGULATOR

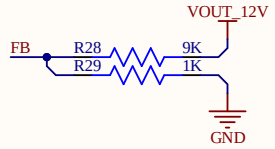


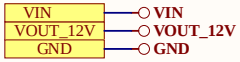
VIN=36V
VOUT=12V, where VOUT= 1.235(1+R2/R1)*
For 12V output, R2=9K and R1=1K
(Datasheet pg.3)

For IOUT=2A, RCS=75mA, (RCS=0.155/IOUT)

Reference: XL4001 Datasheet

FEEDBACK NETWORK

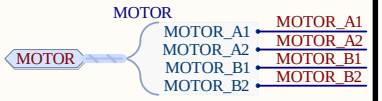




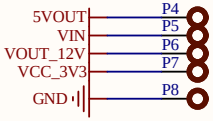
CONTROL SIGNALS



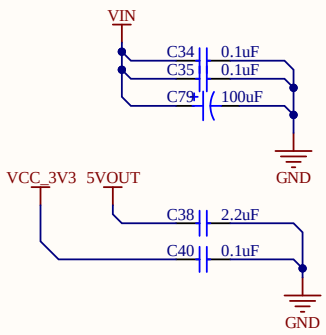
MOTOR OUT



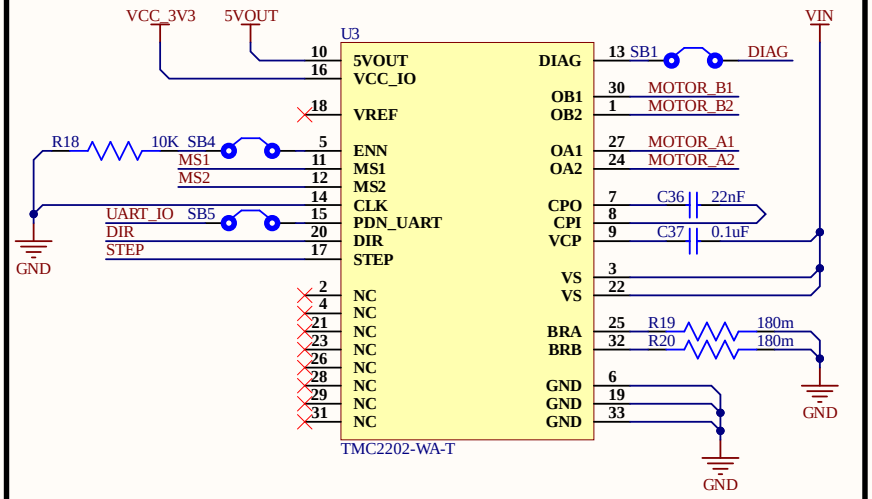
TEST POINTS



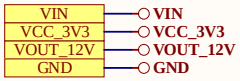
DECOUPLING CAPACITORS



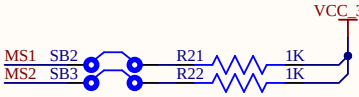
STEPPER DRIVER



POWER SIGNALS



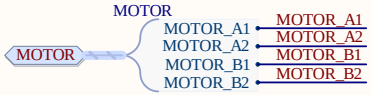
MANNUAL MICROSTEPPING



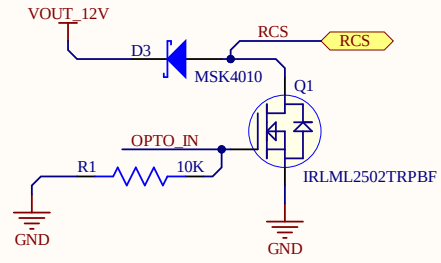
CONTROL SIGNALS



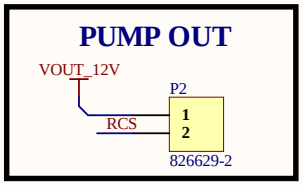
MOTOR OUT



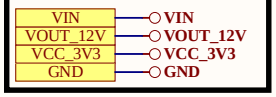
PUMP CONTROL



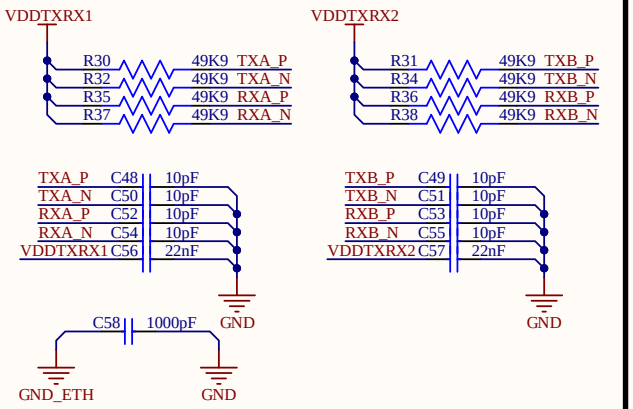
PUMP OUT



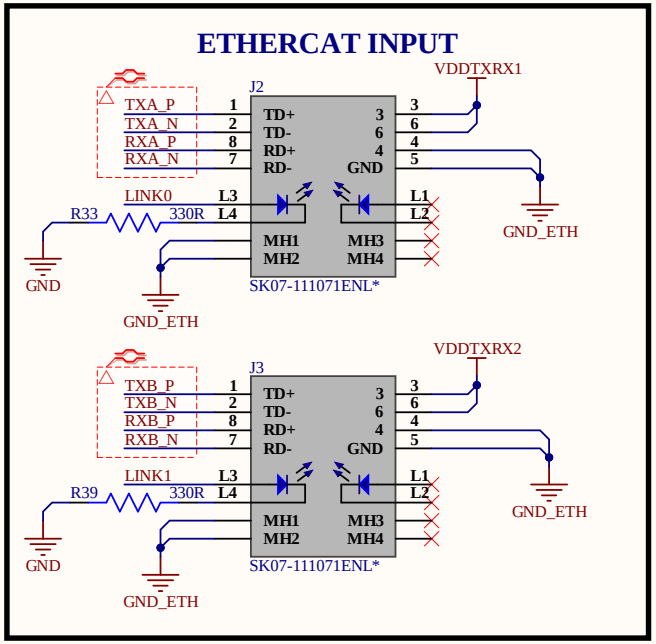
POWER SIGNALS



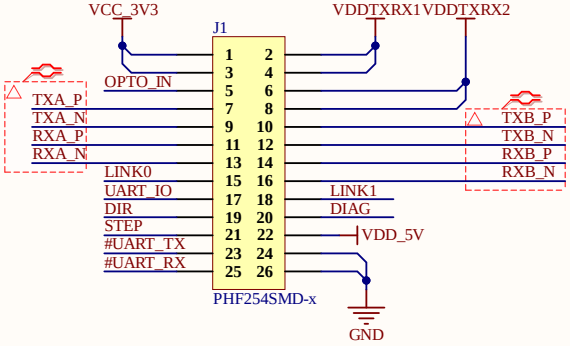
SUPPORTING CIRCUITRY



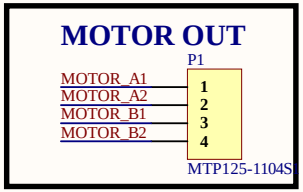
ETHERCAT INPUT



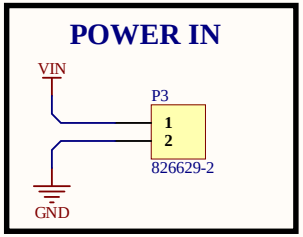
BOARD TO BOARD HEADER



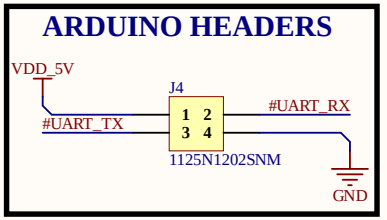
MOTOR OUT

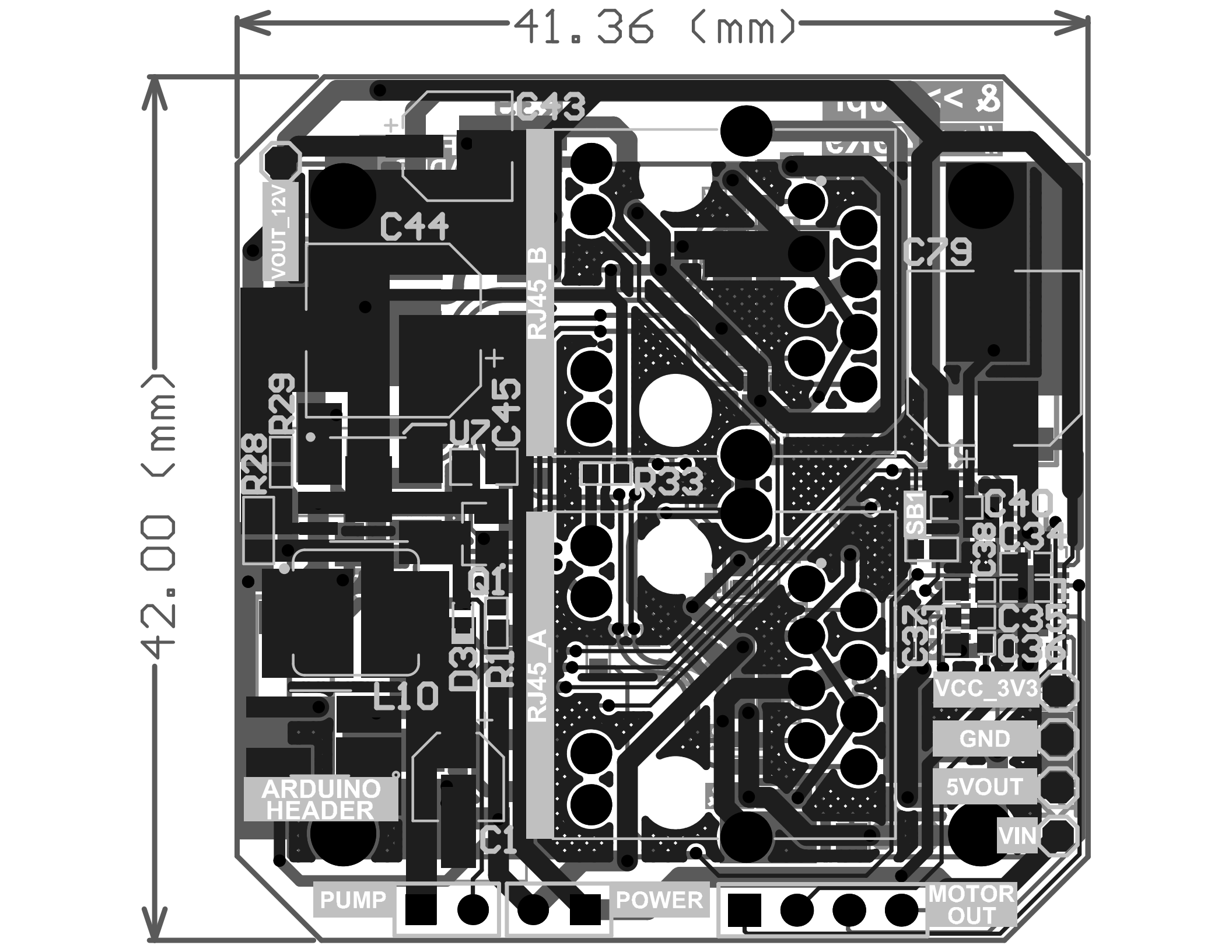


POWER IN



ARDUINO HEADERS




41.36 (mm)
42.00 (mm)
C43
C44
VOUT_12V
C79
RJ45_B
R29
R28
U7
C45
R33
SB1
C38
C40
C34
C37
C35
C36
D3
D1
R1
L10
RJ45_A
C1
VCC_3V3
GND
5VOUT
VIN
ARDUINO HEADER
PUMP
POWER
MOTOR OUT